
BPM2_NB_ESOFT QUIZ 5_S2026

of Questions: 30

Question #: 1

A 67-year-old man comes to the physician because he is unable to close his right eye. Physical examination shows left eyelid is able to close without any problem. Which of the following nerve lesions most likely will lead to inability to close the eye?

- A. CN III
- ✓B. CN VII
- C. CN V3
- D. CN II
- E. Cervical Sympathetic chain

Rationale: CN III would give a total ptosis; CN VII will lead to orbicularis oculi affected and inability to close the eye; Mandibular nerve has no relation with the eye muscles and CN II is special sense nerve. Sympathetic chain in the neck if affected could lead to pseudoptosis.

Item Description: CN III nerve lesion: ptosis

Question #: 2

A 43-year-old patient comes to the physician because the tongue deviates toward the right side, after endarterectomy procedure. Physical examination shows the sensory innervation of the head and neck soft tissues, as well as the laryngeal muscles, appears unaffected. Which of the following structures is most likely damaged by endarterectomy procedure?

- ✓A. Hypoglossal nerve and hypoglossal canal
- B. Mandibular nerve and foramen ovale

- C. Facial nerve and internal acoustic meatus
- D. Glossopharyngeal nerve and jugular foramen
- E. Facial nerve and stylomastoid foramen

Rationale: You should immediately think of CN XII if your tongue is affected. The C1 ventral ramus joins CN XII as it travels inferolaterally into the carotid sheath space. Because the laryngeal muscles are not affected, CN X involvement can be ruled out as a possibility.

Item Description: Hypoglossal nerve and hypoglossal canal

Question #: 3

A 24-year-old man come to physicians office because he collapsed to the ground after a ball hit right side of his head and was unconcious for about four minutes. Once concious he expericend with a severe headache and dousy. Physical examination shows bruise on right temple and dilated right pupil. Radiological examination shows a fracture of the squamous part of the right temporal bone. and an epidural hematoma. Which of the following structures is most likely damaged?

- ✓A. Middle meningeal artery
- B. Accessory meningeal artery
- C. Infraorbital artery
- D. Sphenopalatine artery

Rationale: The middle meningeal artery originates in the infratemporal fossa and ascends through the foramen spinosum into the middle cranial fossa of the cranial cavity. It supplies the dura mater and the skull of the lateral cranial vault. An epidural hematoma is caused by a bleed in the middle meningeal artery.

Item Description: Middle meningeal artery

Question #: 4

A 42-year-old woman comes to the physician because of a 2-day history of blurred vision. Her temperature is 100.4°F (38°C). Physical examination shows no structural abnormalities. The physician suspects uveitis (inflammation of the uveal tract). A normal photomicrograph of the anterior portion of the affected structure is shown. Which of the following is a characteristic feature of the circumscribed structure?

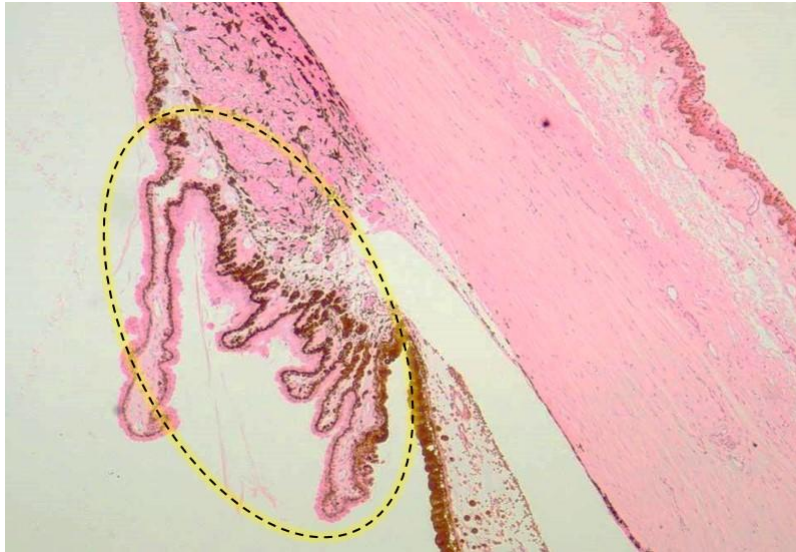
- ✓A. Has epithelial cells that produce and secretes aqueous humor
- B. Contains smooth muscles that reduces the size of the pupil
- C. Contains smooth muscles that increases the size of the pupil

D. Possesses the Meibomian glands

E. Synthesizes and secretes the vitreous humor

Rationale: The circumscribed structure is the ciliary body. The ciliary body is made up of smooth muscles and ciliary processes. The smooth muscles work in tandem to alter the shape of the lens, not the pupils. Pupillary constriction and dilation are achieved by the constrictor pupillae and the dilator pupillae respectively, both of which are found in the iris, not the ciliary body. The ciliary processes of the ciliary body produce both the zonule fibers and the aqueous humor found in the anterior and posterior chambers of the eyes. Meibomian glands are part of the eyelids.

Attachment:



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Item Description: HCB.1338 - Ciliary body

Question #: 5

A 24-year-old woman comes to the physician because of a 1-hour history of severe upper abdominal pain. She has a 2-year-history of recurrent heartburn. Endoscopy is done and a biopsy is obtained from the lower end of esophagus. Which of the following epithelia most likely normally lines this region?

- A. Simple cuboidal
- B. Simple squamous
- C. Pseudostratified
- D. Stratified squamous keratinized
- ✓E. Stratified squamous non-keratinized

Rationale: The esophagus is an organ of the gastrointestinal tract that serves as a conduit for food and other substances from the oropharynx to the stomach. Its lumen is lined by the non-keratinized stratified squamous epithelium of the mucosa.

Question #: 6

A 70-year-old man is brought to the physician because of a 4-month history of a progressive loss of vision in both eyes. Physical examination shows reduced visual acuity in both eyes. A slit lamp examination of both eyes shows inclusion bodies within the cells shown at the tip of the arrows in the attached photomicrograph. Which of the following best describes a function of these cells?

- A. Maintenance of the shape of the eyes
- B. Prevent posterior spread of infection
- ✓C. Maintain optimal hydration of the corneal stroma
- D. Absorption of stray light
- E. Adjustment of pupil diameter
- F. Maintenance of the shape of the lens

Rationale: Transparency of the cornea requires precise regulation of the water content of the stroma. The corneal endothelium functions in the metabolic exchange between the cornea and aqueous humor via the Na⁺/K⁺ ATPase pump.

Option a - sclera

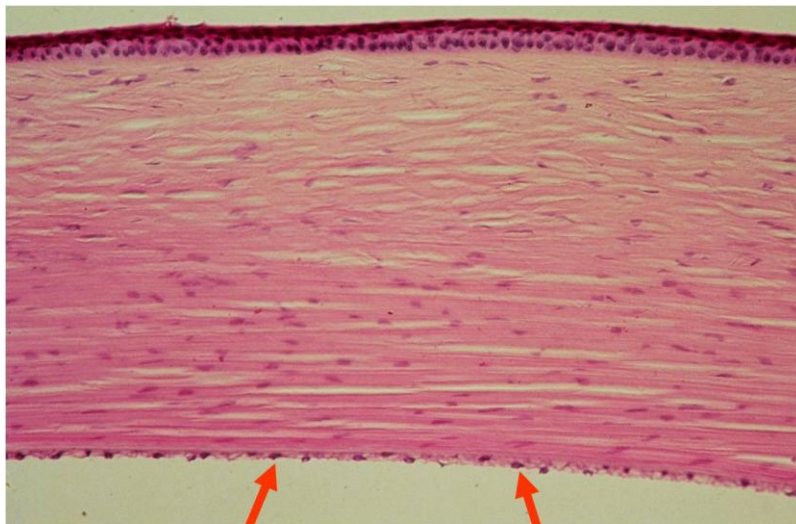
Option B - Bowman's membrane

Option D - Iris, retinal pigmented epithelium

Option E - Iris

Option F - ciliary body

Attachment:



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Question #: 7

A 45-year-old man comes to the physician because of a 3-week history of yellowish discoloration of the eyes. Physical examination shows jaundice and ascites. Laboratory studies are shown. Which of the following best explains the pathogenesis of this disorder?

- A. Increased neutrophil elastase activity
- B. Decreased dietary copper absorption
- ✓C. Increased dietary iron absorption
- D. Decreased biliary copper excretion
- E. Mutation in membrane-bound gated-chloride ion channel
- F. Decreased iron availability for erythropoiesis

Rationale: The vignette shows a patient with hemochromatosis. The liver is affected due to chronic iron accumulation. Elevated serum ferritin levels are a characteristic laboratory finding.

Other organs affected are the skin (bronzing of skin), beta cells of pancreas (secondary diabetes), joints (joint pain) and heart (cardiomyopathy). These patients have unregulated and excessive dietary iron absorption due to impaired intestinal mucosal iron sensing mechanisms.

An important differential diagnosis is Wilson disease (copper overload) which is characterized by liver damage, Kayser Fleischer ring in cornea and deposition of copper in the basal ganglia of brain. Serum ceruloplasmin levels are very low. Characterized by a defect in incorporation of copper to ceruloplasmin and impaired excretion of copper in bile. Urinary copper excretion is usually elevated in these patients. Alpha 1- antitrypsin deficiency can affect the lungs (COPD) and liver (in some patients). The emphysema in these patients is due to excessive neutrophil elastase activity.

Menke disease is characterized by poor absorption of dietary copper. Typically presents in infancy. Characterized by kinky hair, hypopigmentation and aneurysms. The liver is NOT affected in Menke disease.

Patients with cystic fibrosis have repeated respiratory infections and malabsorption due to block of exocrine pancreatic secretions by the viscid secretions. The liver is NOT commonly affected. Cystic fibrosis is characterized by a mutation in a transmembrane gated-chloride ion channel. An important lab finding is elevated sweat chloride levels.

Body iron content is low in iron deficiency anemia: This leads to anemia due to reduced rates of erythropoiesis. Iron deficiency anemia may be due to low dietary iron content or due to excessive blood loss. Liver function is not affected in iron deficiency.

Attachment:

	Patient	Normal range
Alanine aminotransferase (ALT)	180	40-60
Aspartate aminotransferase (AST)	250	40-60
Alkaline phosphatase (ALP)	140	50-70
Gamma glutamyl transferase (GGT)	130	50-70
Serum ferritin	340	20-250 ng/mL

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Item Description: Hemochromatosis

Question #: 8

A 46-year-old woman comes to the physician because of a 3-month history of yellowish discoloration of the eyes and skin. Laboratory and imaging studies show obstruction of the common bile duct by gallstones. Which of the following laboratory tests best reflects this patient's condition?

- A. I
- B. II
- C. III
- D. IV
- ✓E. V

Rationale: The vignette describes a patient with posthepatic (obstructive) jaundice. Serum ALP and GGT levels are usually elevated in this type of jaundice. The conjugated bilirubin (direct bilirubin) is elevated and is found in urine. Urobilinogen in urine is reduced or absent due to the obstruction. In hemolytic jaundice, unconjugated (indirect) bilirubin is elevated, bilirubin is absent in urine and urine urobilinogen levels are raised. In hepatic jaundice, direct and indirect bilirubin levels are increased, bilirubin is present in urine.

Attachment:

	Type of serum bilirubin	Urine bilirubin	Urine urobilinogen
I	Indirect and direct	Present	Decreased
II	Unconjugated and conjugated	Absent	Increased
III	Unconjugated	Absent	Increased
IV	Direct	Absent	Absent
V	Conjugated	Present	Absent

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Item Description: Posthepatic jaundice

Question #: 9

A 9-month-old girl is brought to the emergency room because of a 2-hour history of drowsiness and refusal of feeds. Two days earlier, she had fever, vomiting and diarrhea. Urinalysis shows octanoyl (C-8) and decanoyl (C-10) carnitine derivatives. Which of the following is the most likely diagnosis?

- A. von Gierke disease
- B. McArdle syndrome
- C. Myopathic carnitine deficiency
- ✓D. Medium chain acyl CoA dehydrogenase deficiency
- E. Defective peroxisomal biogenesis
- F. Maple syrup urine disease

Rationale: The vignette describes a child with MCAD deficiency. Note the hypoglycemia, hypoketosis and increased levels of medium chain acylcarnitines.

Attachment:

ANALYTE	Patient	Reference Range
Glucose (mg/dL)	40	70 - 110
Acetoacetate (mg/dL)	0.1	0.2 - 1.0

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Item Description: MCADD

Question #: 10

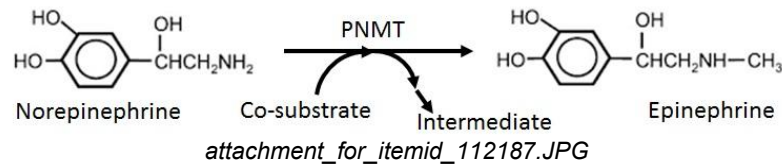
An investigator is studying catecholamine neurotransmitters and focuses on the formation of epinephrine from norepinephrine. It is found that an intermediate labeled in the figure is formed that must be removed. If metabolism of this intermediate is blocked, a compound will accumulate which will result in associated pathologic changes. A figure depicting the catalyzed reaction is shown. Which of the following intermediate molecules is most likely formed when this reaction takes place?

- A. Carbon monoxide
- B. Phenylpyruvic acid
- ✓C. Homocysteine
- D. Nitric oxide
- E. Methylmalonic acid
- F. Delta-aminolevulinate
- G. Citrulline

Rationale: S-adenosyl methionine (SAM) is the methyl donor in the PNMT reaction. SAM is converted to S-Adenosylhomocysteine (SAH). SAH is later converted to Homocysteine.

The further fates of homocysteine are that it can be converted to cysteine (requires PLP) or it can be recycled back to methionine (Requires methyl tetrahydrofolate and vitamin B12).

Attachment:



Item Description: QUIZ - formation of catecholamines

Question #: 11

A 61-year-old man comes to the physician because of a 1-week history of shortness of breath and productive cough. He has smoked 1 pack of cigarettes daily for 40 years. Physical examination shows diffuse rhonchi and wheezing. A chest x-ray and pulmonary function tests confirm the diagnosis of an acute exacerbation of chronic obstructive lung disease. He is administered a drug that stimulates adrenergic receptors in the bronchial smooth muscles producing bronchodilation. Which of the following intracellular mediators is most likely increased in the bronchial smooth muscles of this patient?

- ✓A. cAMP
- B. cGMP
- C. DAG
- D. Protein kinase C
- E. Nitric oxide

Rationale: Stimulation of beta2 adrenergic receptors cause relaxation of the smooth muscles and bronchodilation. Drugs that stimulate these receptors have similar effect. Stimulation of adrenergic receptors activates Gs-protein mediated adenylyl cyclase which increases cAMP in the smooth muscles. Increase in cAMP causes relaxation of the bronchial sm muscles.

Item Description: PHYS.0044 Adrenergic receptors Cellular signaling

Question #: 12

A 29-year-old man comes to the physician because of a 3-month history of headaches and decreased libido. His vital signs are within normal limits. Physical examination shows bitemporal hemianopsia, bilateral gynecomastia, and scanty hairs on face, groin and axilla. Laboratory studies show reduced serum levels of testosterone. MRI of the brain shows enlargement of the Sella turcica. Which of the following hormones is most likely inhibited in this patient?

- A. Thyroid stimulating hormone
- B. Growth hormone
- C. Prolactin
- ✓D. Gonadotropin releasing hormone
- E. Oxytocin

Rationale: This is a case of prolactinoma where there is secretion of prolactin from the adenoma and this causes a rise in prolactin levels which leads to all the symptoms described. Prolactin inhibits the secretion of GnRH. This results in low LH, FSH and testosterone.

Item Description: PHYS.1017. Prolactinoma laboratory findings

Question #: 13

A 43-year-old man comes to the physician because of a 3-month history of recurrent episodes of headaches, blurry vision, and exertional dyspnea. His blood pressure is 156/96 mm Hg; other vital signs are within normal limits. Physical examination shows a loud S2 in aortic area. Laboratory studies show elevated serum sodium, reduced serum potassium, and elevated serum bicarbonate. CT scan of the abdomen shows a mass in the right adrenal gland. He is diagnosed with an aldosterone-secreting tumor. Increased activity of which of the following renal tubular transport mechanisms is most likely in this patient?

- A. Sodium-Glucose secondary active transport in the proximal convoluted tubules
- B. Sodium-Potassium-Chloride transporter in the thick ascending limb of loop of Henle
- C. Sodium-Amino acid transporter in the proximal convoluted tubules
- ✓D. Epithelial Sodium channels in the collecting duct
- E. Sodium-chloride transporter in the distal convoluted tubules

Rationale: This patient secretes large amount of aldosterone because of the tumor. Aldosterone acts on the Epithelial sodium channels and increase sodium reabsorption and potassium excretion in the kidneys.

Item Description: PHYS.1028.Aldosterone action

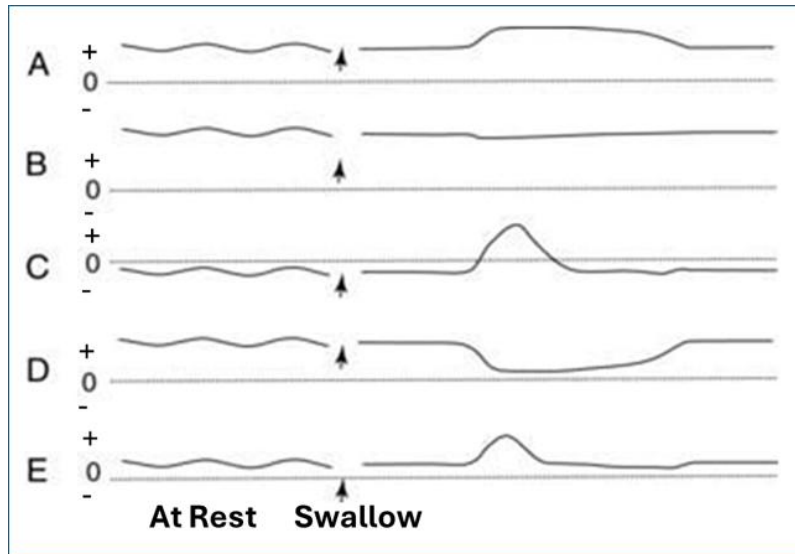
Question #: 14

A 58-year-old man comes to the physician because of a 3-week history of difficulty swallowing solids. He does not have any chronic illnesses. Physical examination shows no abnormalities. An upper GI endoscopy shows no abnormalities. An esophageal manometry shows normal findings. Mid-esophageal manometry findings in normal and diseased conditions are shown in the attached figure. Which of the following manometric recordings best represents this patient?

- A. A
- B. B
- ✓C. C
- D. D
- E. E

Rationale: This patient has a normal esophageal motility. Therefore his manometry finding must be normal. The normal manometric finding for the mid-esophagus (esophageal body) is characterized by a basal subatmospheric pressure and a positive pressure wave caused by the passage of the food bolus. This is best demonstrated in the option C.

Attachment:



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Item Description: PHYS.1101. Esophageal Motility

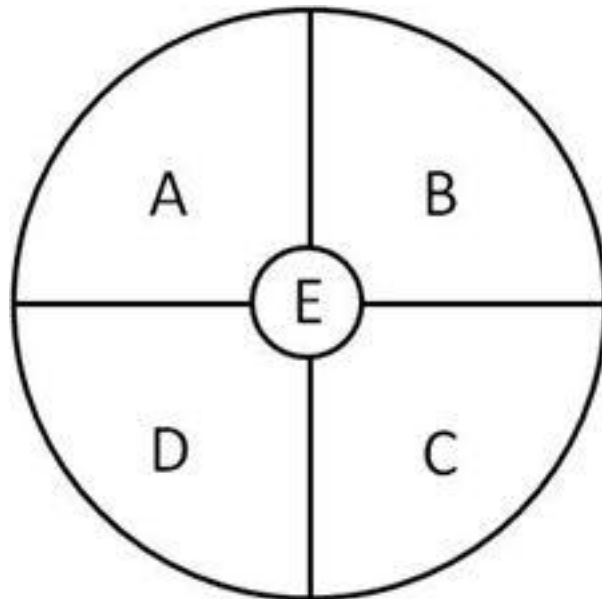
Question #: 15

An 82-year-old woman comes to the physician because of a 9-month history of visual problems. Fundoscopic examination shows retinal mottling consistent with age-related macular degeneration. Which of the following portions of the visual field is most likely disrupted in the patient (see attached figure)?

- A. A
- B. B
- C. C
- D. D
- ✓E. E

Rationale: The macular is a region of the retina that represents the central visual field and has the highest visual acuity. Degeneration of the macula leads to visual loss in the central visual field (region E on the image).

Attachment:



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Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0222 Visual field retina

Question #: 16

A 24-year-old man comes to the physician because of a 3-month history of double-vision and difficulty seeing with his left eye. His left pupil is dilated, and his left eyelid droops. His left eye is tonically deviated laterally and down. Residual function of which of the following extraocular muscles most likely mediates the downward component of the deviation of the eye?

- A. Superior rectus
- B. Inferior rectus
- ✓C. Superior oblique
- D. Inferior oblique
- E. Lateral rectus
- F. Medial rectus

Rationale: The post-traumatic manifestations affecting the left eye suggests injury to CN III. This implies that the lateral rectus and superior oblique muscles may still be exerting residual effects, causing lateral and slightly downward deviation of the eye. The still intact superior oblique muscle could thus cause subtle downward deviation of the eye.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0230 Extraocular function

Question #: 17

A 49-year-old man comes to the physician because of a 1-day history of hearing loss. Weber's test lateralizes to the left. Rinne's test shows AC > BC bilaterally. Which of the following structures is most

likely lesioned?

- A. Right cingulate
- B. Right ossicle chain
- ✓C. Right cochlea
- D. Right caudate
- E. Right insula
- F. Left cingulate
- G. Left ossicle chain
- H. Left cochlea
- I. Left caudate
- J. Left insula

Rationale: The patient has sensorineural deafness, as indicated by bilateral AC>BC. Because the sound lateralizes to the left during Weber's, the injury must be on the right. Of the structures listed, only the cochlea contributes to the auditory pathways.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0239 Auditory pathways

Question #: 18

A 27-year-old man comes to the physician because of a 2-week history of hearing problems. He says that a fire-cracker had exploded near his left ear. Tuning fork tests show sensorineural hearing loss affecting the left ear and a diagnosis of acquired hearing loss is made. Which of the following findings during Weber's test is most likely?

- A. Lateralization to the left ear
- ✓B. Lateralization to the right ear
- C. Bilaterally normal hearing
- D. BC = AC = 0 bilaterally
- E. BC >AC for the right ear

Rationale: Sensorineural hearing on the left ought to yield lateralization of sound to the unaffected ear (the right ear in this case). AC ought to be greater than BC for both ears, but total times ought to be reduced for the left ear.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0239 Auditory pathways

Question #: 19

A 47-year-old man comes to the physician because of a 5-day history of headaches. Assessment of pupillary light reflexes shows a loss of pupillary

constriction bilaterally with presentation of light to his right eye. Both pupils constrict with the presentation of light to his left eye. Which of the following structures is most likely lesioned in this patient?

- A. Left CN III
- B. Right CN III
- C. Left CN II
- ✓D. Right CN II
- E. Left CN V
- F. Right CN V

Rationale: Because both pupils constrict with presentation of light to the left eye, it is apparent that the sensory and motor limbs of the reflex are intact (i.e., left CN II and right and left CN III). The failure of pupillary constriction with the presentation of light to the right eye, however, demonstrates a sensory loss (i.e., right CN II).

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0210 Pupillary light reflex

Question #: 20

A 57-year-old man is brought to the emergency department because of a 2-hour history of problems seeing. Confrontation testing shows a visual field defect. Imaging studies show an infarction involving the occipital branch of the right posterior cerebral artery. Which of the following visual field defects is most likely in this patient?

- A. Bitemporal hemianopia
- B. Left superior homonymous quadrantanopia
- C. Right superior homonymous quadrantanopia
- D. Left inferior homonymous quadrantanopia
- E. Right inferior homonymous quadrantanopia
- F. Bitemporal hemianopia with macular sparing
- G. Left homonymous hemianopia
- H. Right homonymous hemianopia
- ✓I. Left homonymous hemianopia with macular sparing
- J. Right homonymous hemianopia with macular sparing

Rationale: Much of the primary visual cortex on the right could be affected by this infarction, yielding a left homonymous hemianopia. However, it is common in such cases for the middle cerebral artery to lend some blood to the otherwise deprived occipital pole, which can therefore result in sparing of macular vision.

Item Description: SOM.MK.I.BPM2.1.NB.4.NSCI.0228 Visual pathway lesions

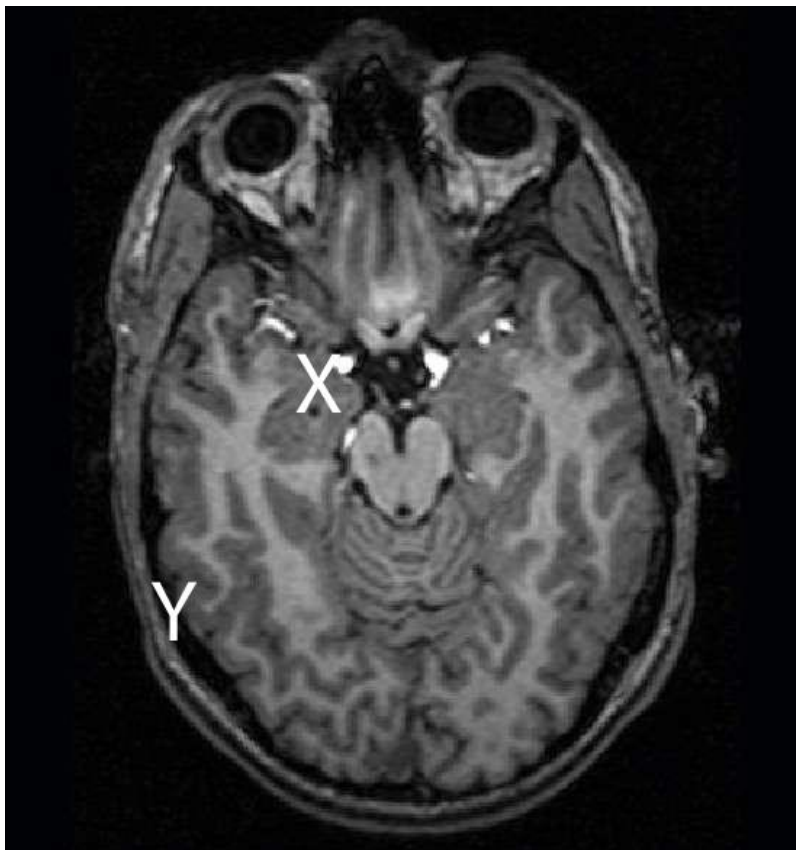
Question #: 21

A 52-year-old man is brought to the emergency department because of a 2-hour history of a traumatic head wound. Imaging studies show that a fast-moving fragment of metal entered the skull at Site Y and lodged at Site X. Which of the following trajectories is most likely followed by the fragment as it passes through the brain?

- ✓A. Anteromedial
- B. Posteromedial
- C. Anterolateral
- D. Posterolateral
- E. Dorsomedial
- F. Ventromedial
- G. Dorsolateral
- H. Ventrolateral

Rationale: Both Y and X correspond to the telencephalon. X lies anterior to Y (X is closer to the face). X also lies closer to the midline than does Y. Hence, the fragment has traveled anteromedially.

Attachment:



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Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0009 Neuroimaging anatomical vectors

Question #: 22

A 47-year old man is brought to the emergency department because of a 3-hour history of nausea and a fainting spell. Imaging studies show hypoperfusion in a small localized area of the brain. An infarction is suspected. Which of the following cells will most likely proliferate to form a scar in the injured area?

- A. Schwann cells
- B. Microglia
- C. Oligodendrocytes
- ✓D. Astrocytes
- E. Ependymocytes

Rationale: Astrocytes are the glial cells that form scars in the central nervous system.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0017 Glial cells

Question #: 23

A 49-year-old man comes to the physician because of a 1-day history of hearing loss. Weber's test lateralizes to the left. Rinne's test shows AC > BC bilaterally. Which of the following structures is most likely lesioned?

- A. Right cingulate
- B. Right ossicle chain
- ✓C. Right CN VIII
- D. Right caudate
- E. Right insula
- F. Left cingulate
- G. Left ossicle chain
- H. Left CN VIII
- I. Left caudate
- J. Left insula

Rationale: The patient has sensorineural deafness, as indicated by bilateral AC > BC. Because the sound lateralizes to the left during Weber's, the injury must be on the right. Of the structures listed, only CN VIII contributes to the auditory neural pathways.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0239 Auditory pathways

Question #: 24

A 44-year-old man comes to the physician because of an 8-hour history of clumsiness. Imaging studies show a lesion overlapping with the left cerebocerebellum. Which of the following clinical manifestations

is most likely in this patient?

- A. Left-sided tingling
- B. Right-sided tingling
- ✓C. Left dysdiadochokinesia
- D. Right dysdiadochokinesia
- E. Left resting tremor
- F. Right resting tremor

Rationale: Dysdiadochokinesia on the left is most likely, as the posterior lobe enabled coordination of the ipsilateral body. Somatosensory manifestations ought not to be present. Resting tremor is more closely associated with Parkinson disease.

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 25

A 68-year-old woman comes to the physician because of a 6-month history of blurry vision. Fundoscopic studies show bilateral macular lesions. Which of the following processes is most likely?

- A. Loss of lenticular elasticity
- B. Occipital stroke
- ✓C. Neurodegeneration
- D. Demyelination
- E. Diabetic optic neuritis

Rationale: A degenerative process affecting the macula is most likely. Bilateral stroke affecting the occipital pole could yield similar visual impairments, but lesions of the retina would not likely also be seen.

Item Description: SOM.MK.I.BPM2.4.NB.4.NSCI.0215 Causes of visual impairment

Question #: 26

A 78-year-old man comes to the physician because of a 4-day history of clumsiness of his right arm and leg. He also has difficulty maintaining his balance in the dark. Physical examination shows diminished touch, vibration and conscious proprioception for the right arm and leg. Imaging studies show a spinal infarction. Which of the following arterial regions is most likely implicated?

- A. Anterior spinal artery (cervical level)

- B. Left posterior spinal artery (thoracic level)
- C. Left posterior spinal artery (cervical level)
- D. Right posterior spinal artery (thoracic level)
- ✓E. Right posterior spinal artery (cervical level)

Rationale: The findings indicate damage to the right posterior columns. This area is supplied by the right posterior spinal artery. Because both the arm and the leg are affected, the lesion must be situated rostrally in the cord.

Item Description: SOM.MK.I.BPM2.5.NB.5.NSCI.0364 Spinal lesions and sensory loss

Question #: 27

A 27-year-old man comes to the physician because of a 6-month history of short, unilateral attacks of severe, shooting facial pain, that feel "like an electric shock." Each pain attack lasts only a few seconds, but they can cluster, followed by pain-free intervals lasting for several weeks. The pain attacks can be triggered by light touch near the left corner of the mouth and by eating or talking. MRI of the head is normal. Dental conditions are normal. The pain is refractory to pharmacological intervention. Surgical ablation of pseudounipolar cells is considered. Which of the following structures is most closely associated with the proximal root of the surgically targeted cell bodies?

- A. Cerebrum
- B. Diencephalon
- C. Midbrain
- ✓D. Pons
- E. Medulla

Rationale: The pain originates in the face, in the dermatomes innervated by the trigeminal nerve. Cell bodies forming the trigeminal ganglion sit lateral to the pons.

Item Description: SOM.MK.IV.BPM2.4.NB.4.NSCI.0126 Pain management

Question #: 28

A 54-year-old man comes to the physician because of an 8-hour history of clumsiness. Imaging studies show a lesion overlapping with the right lateral hemisphere. Which of the following clinical manifestations is most likely?

- A. Right-sided tingling
- B. Left-sided tingling
- C. Left dysdiadochokinesia
- ✓D. Right dysdiadochokinesia
- E. Left resting tremor

F. Right resting tremor

Rationale: Dysdiadochokinesia on the left is most likely, as the lateral hemisphere enables coordination of the ipsilateral body. Somatosensory manifestations ought not to be present. Resting tremor is more closely associated with Parkinson disease.

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 29

A 21-year-old man is brought to the emergency department because of a 40-minute history of a traumatic injury following a fragment of metal penetration to his back. CT scans of his trunk show a right spinal hemisection at T10. Which of the following manifestations is most likely?

- ☒ A. Loss of nociception of the left leg
- ☐ B. Loss of proprioception of the left leg
- ☐ C. Weakness of the left leg
- ☐ D. Loss of nociception of left dermatome T10
- ☐ E. Loss of nociception of the right arm

Rationale: The hemisection of the spinal cord predicts the onset of Brown-Sequard syndrome. Given that the lesion occurs at spinal T10, the arms should be unaffected. Nociception ought to be lost contralaterally below the level of the lesion, whereas proprioception ought to be lost ipsilateral to the lesion. Weakness of the leg should arise ipsilateral to the injury.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0119 Use knowledge of the anatomical and functional organization of the somatosensory system to identify and localize the clinical correlations Shingles (herpes zoster), Brown-Sequard syndrome and Syringomyelia

Question #: 30

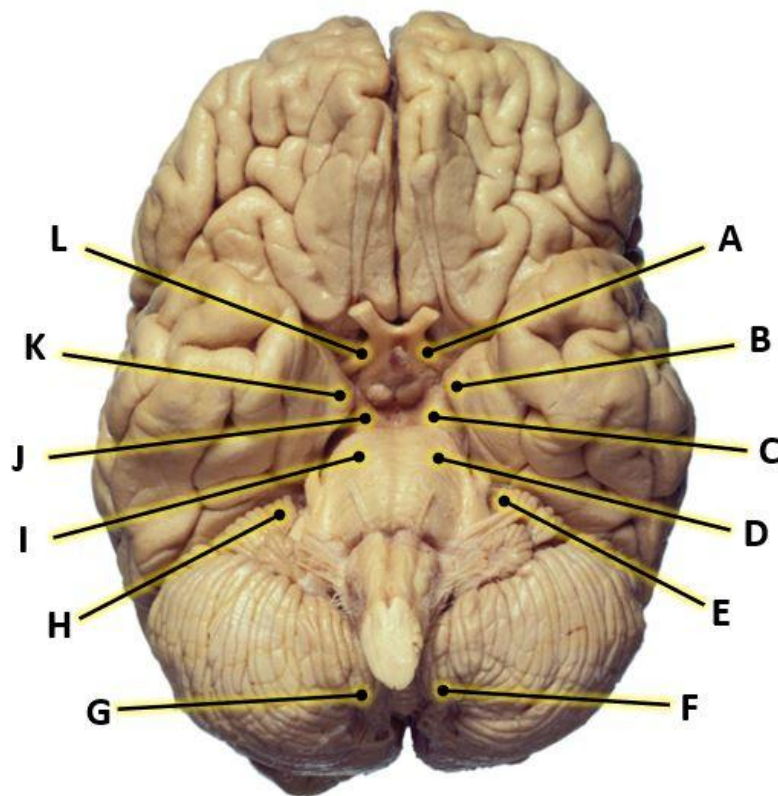
A 54-year-old man is brought to the emergency department because of weakness on the right side of his body, including upper and lower extremities that developed over the past few months. Neurologic examination confirms a right hemiplegia. His left eye pupil is abnormally dilated and responds sluggishly to light. Left eye movements are deficient. Herniation of which of the structures marked in the brain image attached through the tentorial notch is the most likely cause of the pathological findings in this patient?

- ☐ A. A
- ☒ B. B
- ☐ C. C
- ☐ D. D

- E. E
- F. F
- G. G
- H. H
- I. I
- J. J
- K. K
- L. L

Rationale: The uncus, which is part of the temporal lobe located in the middle cranial fossa, is the supratentorial structure most likely displaced through the tentorial notch, in case of increased supratentorial pressure, causing ipsilateral oculomotor palsy and contralateral hemiplegia. Correct answer: K

Attachment:



attachment_for_itemid_345331.JPG

Item Description: Location of the uncus, uncal herniation